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R&D and Productivity: Estimating Endogenous Productivity

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1 Big picture

- **Question.** How should we estimate the effect of **R&D on productivity** when productivity is unobserved, input choices respond to it, and the R&D process may be **uncertain, nonlinear, and heterogeneous across firms**?
- **Setting.** The paper develops a **dynamic investment model** in which firms invest both in **physical capital** and in **knowledge through R&D**, and applies it to an unbalanced panel of about **1,870 Spanish manufacturing firms in 9 industries** during the **1990s**.
- **Core challenge.** Standard productivity estimation faces simultaneity: firms choose labour and materials after observing productivity, so inputs are correlated with the unobserved productivity shock. At the same time, the classic **knowledge capital model** is restrictive because it treats the stock of knowledge as a deterministic accumulation of past R&D expenditures with simple depreciation.
- **Main methodological contribution.**
 1. Productivity follows a **controlled Markov process** that depends on both lagged productivity and lagged R&D.
 2. Instead of constructing knowledge capital directly, the paper treats productivity as **latent** and recovers it from the firm's **static input choices**.
 3. The paper derives a **parametric inversion** of labour demand, which yields a simple estimator for both the **production function** and the **law of motion for productivity**.
- **Why it matters.** The framework allows R&D to affect productivity in a way that is **state dependent, stochastic, and firm specific**. This lets the authors study not just an average elasticity, but also the full distribution of returns, persistence, and productivity growth.
- **Main empirical question.** How much does R&D account for differences in productivity across firms and the evolution of productivity over time, and is the R&D–productivity link fundamentally **linear and predictable** or **nonlinear and uncertain**?
- **Main empirical results.**
 - An **exogenous** productivity process is rejected in **all industries**.
 - A **separable** R&D process is rejected in **8 of 9 industries at the 5% level** and in **all industries at the 10% level**.
 - The unpredictable component of productivity accounts for roughly **25% to 75%** of total productivity variance, depending on the industry.
 - Firms that perform R&D tend to have **higher expected productivity** than non-performers.
 - R&D-performing firms contribute between about **65% and 90%** of productivity growth in industries with medium or high innovative activity.
 - The average elasticity of output with respect to R&D expenditures is close to **0.015**, but it varies substantially across industries and across firms within industries.
 - The **net rate of return to R&D** is often around **40%** and is frequently **larger than the net return to physical capital**.

- Productivity is **less persistent and more fluid** than what the knowledge capital model suggests.
- **Economic takeaway.** R&D is a major engine of firm productivity, but it is not a smooth deterministic accumulation process. Much of R&D goes into **maintaining existing productivity**, the payoff is **uncertain**, and the effect of current R&D depends strongly on the firm's current productivity level.

2 Data and measurement (key objects)

2.1 Survey, sample, and coverage

- **Data source.** The data come from the *Encuesta Sobre Estrategias Empresariales* (ESEE), a firm-level survey of Spanish manufacturing sponsored by the Ministry of Industry.
- **Unit of observation.** The survey unit is the **firm**, not the plant or establishment.
- **Sampling design.**
 - In 1990, **5%** of firms with up to **200 workers** were randomly sampled by industry and size strata.
 - All firms with more than **200 workers** were invited to participate, and around **70%** responded.
 - New firms were added every year to preserve representativeness.
- **Final sample.** The estimation sample includes firms with at least two years of data and covers about **1,870 firms** in **9 industries**. Industry 5 is excluded because of data problems.
- **Panel structure.** The panel is unbalanced. Firms are observed for between **2 and 10 years** during **1990–1999**. Entry and exit are substantial, and many firms remain in the sample only a short time.

2.2 Key variables

- **R&D expenditures.** These include:
 - intramural R&D,
 - outside R&D contracts with laboratories and research centres,
 - imported technology such as patent licensing or technical assistance.

A firm is classified as **performing R&D** whenever it reports positive R&D expenditure.

- **Investment.** Current investment in **equipment goods**, excluding buildings, land, and financial assets, deflated by the investment price index.
- **Capital.** Constructed recursively with a perpetual-inventory-type equation using investment data and an industry-specific depreciation rate.
- **Labour.** Total **hours worked**, computed as number of workers times average hours per worker.
- **Materials.** Intermediate consumption, including raw materials, components, energy, and services, deflated by a firm-specific materials price index.

- **Output.** Value of produced goods and services, equal to sales plus inventory variation, deflated by a firm-specific output price index.
- **Wage.** Hourly labour cost including social security payments divided by total hours worked.
- **Prices.**
 - firm-specific price index for output,
 - firm-specific price index for materials,
 - industry-level price index for investment goods.
- **Demand shifter.** An index of market dynamism based on firms' assessments of whether their markets are contracting, stable, or expanding.

2.3 Descriptive environment and innovative activity

- **Macro background in the data.** The 1990s show rapid **output growth** and intense **physical capital accumulation**, but employment growth is weak or even negative in several industries.
- **Price growth.** Output price growth is moderate, while wage and materials prices rise more strongly.
- **Labour flexibility.** Spain had a high share of temporary workers in the 1990s, and firms appear able to adjust labour at both the extensive and hours margins. The share of temporary workers ranges roughly from **14% to 33%** across industries.
- **High-innovation industries.** Chemical products, agricultural and industrial machinery, and transport equipment have the highest incidence of R&D. In these industries, about **two-thirds** of firms perform R&D at least once.
- **Intermediate-innovation industries.** Metals and metal products, non-metallic minerals, food/drink/tobacco, and textiles/leather/shoes have lower participation, with performer shares below one half.
- **Low-innovation industries.** Timber/furniture and paper/printing have performer shares around **one quarter**.
- **R&D intensity.** Among performing firms, average R&D intensity ranges from roughly **0.7% to 2.7%** depending on the industry.

2.4 Unobservables and timing

- **Unobserved productivity.** Productivity ω_{jt} is observed by the firm but not by the econometrician.
- **Static inputs.** Labour and materials are treated as **static inputs**: they are chosen after the firm observes current productivity.
- **Dynamic inputs.** Physical capital and knowledge investment are **dynamic decisions**.
- **Timing.**

- At time $t - 1$, the firm chooses investment in physical capital and R&D.
- At time t , that investment becomes productive, productivity is realized, and the firm chooses labour and materials.

3 Models and empirical specifications (all equations + interpretation)

3.1 Dynamic investment problem

A firm invests in both physical capital and knowledge. Physical capital evolves according to

$$K_{jt} = (1 - \delta)K_{j,t-1} + I_{j,t-1}.$$

Interpretation. Investment chosen in period $t - 1$ becomes productive in period t , so capital is the dynamic state variable among standard inputs.

The production function is Cobb–Douglas:

$$y_{jt} = \beta_0 + \beta_t t + \beta_k k_{jt} + \beta_l l_{jt} + \beta_m m_{jt} + \omega_{jt} + e_{jt}.$$

- y_{jt} : log output
- k_{jt} : log physical capital
- l_{jt} : log labour
- m_{jt} : log materials
- ω_{jt} : latent productivity observed by the firm
- e_{jt} : transitory shock, uncorrelated over time and across firms

Interpretation. The firm's output depends on conventional inputs and a latent productivity component. The key econometric problem is that labour and materials respond to ω_{jt} .

The firm's Bellman equation is

$$V(s_{jt}) = \max_{i_{jt}, r_{jt}} \left\{ \Pi(s_{jt}) - C_i(i_{jt}) - C_r(r_{jt}) + \frac{1}{1 + \rho} \mathbb{E}[V(s_{j,t+1}) \mid s_{jt}, i_{jt}, r_{jt}] \right\},$$

where the state vector is

$$s_{jt} = (t, k_{jt}, \omega_{jt}, w_{jt}, p_{jt}^M, d_{jt}).$$

Interpretation. The state includes time, capital, productivity, the wage, the price of materials, and a demand shifter. R&D is a forward-looking investment decision because it shifts future productivity.

3.2 Controlled Markov process for productivity

The central modelling step is to endogenize productivity as

$$\omega_{jt} = \mathbb{E}[\omega_{jt} \mid \omega_{j,t-1}, r_{j,t-1}] + \xi_{jt} = g(\omega_{j,t-1}, r_{j,t-1}) + \xi_{jt}.$$

Interpretation.

- $g(\omega_{j,t-1}, r_{j,t-1})$ is **expected productivity** conditional on current productivity and R&D.
- ξ_{jt} is the **productivity innovation**: the unpredictable component that captures uncertainty in both general productivity evolution and the R&D process.

This is the key departure from models that assume exogenous productivity dynamics.

3.3 Static input demands and parametric inversion

Because labour and materials are chosen after the firm observes current productivity, they can be used to invert for ω_{jt} . Under downward-sloping demand and Cobb–Douglas technology, labour demand is

$$l(s_{jt}) = \frac{1}{1 - \beta_l - \beta_m} \left[\beta_0 + (1 - \beta_m) \ln \beta_l + \beta_m \ln \beta_m + \mu + \beta_t t + \beta_k k_{jt} + \omega_{jt} \right. \\ \left. - (1 - \beta_m)(w_{jt} - p_{jt}) - \beta_m(p_{jt}^M - p_{jt}) + \ln \left(1 - \frac{1}{\eta(p_{jt}, d_{jt})} \right) \right]$$

where $\mu = \ln \mathbb{E}[\exp(e_{jt})]$.

Solving this for productivity yields the inverse labour demand function:

$$h_l(t, k_{jt}, l_{jt}, p_{jt}, w_{jt}, p_{jt}^M, d_{jt}) = \lambda_l - \beta_t t - \beta_k k_{jt} + (1 - \beta_l - \beta_m) l_{jt} \\ + (1 - \beta_m)(w_{jt} - p_{jt}) + \beta_m(p_{jt}^M - p_{jt}) - \ln \left(1 - \frac{1}{\eta(p_{jt}, d_{jt})} \right)$$

with

$$\lambda_l = -\beta_0 - (1 - \beta_m) \ln \beta_l - \beta_m \ln \beta_m - \mu.$$

Interpretation. Current productivity is backed out from an observed static decision. This avoids solving the full dynamic problem as in Olley–Pakes-type approaches.

3.4 Main estimating equation

Substituting the controlled Markov process into the production function and replacing lagged productivity with the lagged inverse labour demand gives

$$y_{jt} = \beta_0 + \beta_t t + \beta_k k_{jt} + \beta_l l_{jt} + \beta_m m_{jt} + g(h_{l,j,t-1}, r_{j,t-1}) + \xi_{jt} + e_{jt}.$$

Interpretation. This is the core equation estimated in the paper. The nonparametric object is only the conditional expectation function $g(\cdot)$; everything else is treated parametrically.

3.5 GMM moment conditions and Markov specification

The estimator uses the moment condition

$$\mathbb{E}[A(z_{jt})(\xi_{jt} + e_{jt})] = 0.$$

Interpretation. Instruments must be orthogonal to the innovation in productivity and the transitory output shock.

The paper allows different productivity laws depending on whether the firm does or does not perform R&D:

$$g(h_{l,j,t-1}, r_{j,t-1}) = \mathbf{1}(R_{j,t-1} = 0) \left[g_{00} + g_{01}(h_{l,j,t-1} - \lambda_l) \right] + \mathbf{1}(R_{j,t-1} > 0) \left[g_{10} + g_{11}(h_{l,j,t-1} - \lambda_l, r_{j,t-1}) \right].$$

Interpretation.

- Firms with zero R&D and firms with positive R&D are allowed to follow different productivity laws.
- The function is estimated flexibly using third-order polynomials.
- This specification directly tests whether the productivity process is exogenous, whether R&D matters only through participation or also through intensity, and whether the process is separable or nonlinear.

3.6 Objects of economic interest: productivity growth, return to R&D, and persistence

Expected productivity growth is

$$\beta_t + \mathbb{E}[\omega_{jt} - \omega_{j,t-1} \mid \omega_{j,t-1}, r_{j,t-1}] = \beta_t + g(\omega_{j,t-1}, r_{j,t-1}) - \omega_{j,t-1}.$$

The paper interprets

$$\frac{\partial g(\omega_{j,t-1}, r_{j,t-1})}{\partial r_{j,t-1}}$$

as the **return to R&D** (the elasticity of output with respect to R&D expenditures), and

$$\frac{\partial g(\omega_{j,t-1}, r_{j,t-1})}{\partial \omega_{j,t-1}}$$

as the **degree of persistence** in productivity.

Interpretation. The model delivers not just one average elasticity, but the entire distribution of returns and persistence across firms and industries.

To study the rate of return to R&D, expected productivity growth is decomposed as

$$\beta_t + g(\omega_{j,t-1}, r_{j,t-1}) - \omega_{j,t-1} = \left[\beta_t + g(\omega_{j,t-1}, r_{j,t-1}) - g(\omega_{j,t-1}, \underline{r}) \right] + \left[g(\omega_{j,t-1}, \underline{r}) - \omega_{j,t-1} \right],$$

where $\underline{r}$ is a negligible amount of R&D.

Interpretation.

- The first bracket is the **gross effect of R&D** on expected productivity growth.
- The second bracket captures the **depreciation or maintenance component** of existing productivity.

Multiplying by value added and dividing by R&D expenditures yields the gross and net rates of return to R&D.

3.7 Comparison benchmark: the knowledge capital model

The classic knowledge capital model is

$$y_{jt} = \beta_0 + \beta_t t + \beta_k k_{jt} + \beta_{l_{jt}} l_{jt} + \beta_m m_{jt} + \varepsilon c_{jt} + e_{jt},$$

with knowledge capital evolving as

$$C_{jt} = (1 - \delta)C_{j,t-1} + R_{j,t-1}.$$

A more flexible Hall–Hayashi/Klette-style version can be written as

$$\omega_{jt} = \sigma \omega_{j,t-1} + \varepsilon(1 - \sigma) \ln(1 + \exp(r_{j,t-1})) + \xi_{jt}.$$

Interpretation. The paper uses these as benchmarks and shows that the simpler knowledge-capital approaches are too restrictive relative to the data.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in standard approaches

- **Knowledge capital models** assume that knowledge accumulation is essentially linear and deterministic once one picks a depreciation rate.
- **Olley–Pakes-type investment inversion** becomes difficult when productivity is endogenous to R&D, because one must prove invertibility from the full dynamic problem and often impose strong index restrictions.

- **Exogenous productivity processes** blur the differences between firms with different innovative activity and mechanically understate the role of R&D.

4.2 What identifies the new approach

- **Static input choices** contain information about current productivity because they are chosen after ω_{jt} is known to the firm.
- **Lagged inverse labour demand** summarizes previously attained productivity in a way that is consistent with the production structure.
- **Lagged R&D** shifts expected future productivity but, under the timing assumptions, is orthogonal to the current innovation ξ_{jt} .
- The model avoids the ACF-style collinearity problem because the nonparametric part depends on the **value** of h_t rather than on the full vector of its arguments.

4.3 Instruments and exclusion logic

- The paper builds instruments from lagged variables and from the closed-form structure of the reduced form.
- Key sources of exogenous variation include:
 - lagged output prices,
 - lagged real wages,
 - lagged real materials prices,
 - lagged labour and materials,
 - current and lagged capital,
 - lagged R&D,
 - lagged demand conditions.
- The current output price is excluded as an instrument because it is endogenous under imperfect competition.

4.4 Empirical credibility of the specification

- **Overidentifying restrictions.** For the main labour inversion, the overidentifying restrictions are not rejected at the 5% level in any industry.
- **Imperfect competition.** Perfect competition is rejected in **7 industries at the 5% level** and in all industries at the 10% level.
- **Parameter restrictions.** Equality between the production-function coefficients and their counterparts in inverse labour demand is rejected only in industries **1** and **7**; the paper argues that remaining discrepancies are plausibly due to measurement error.

- **Alternative inversion.** Using materials instead of labour to recover productivity performs worse, likely because materials are more contaminated by outsourcing, inventories, and measurement error.
- **Permanent vs temporary workers.** An extension that distinguishes permanent and temporary labour gives similar production estimates, so the main conclusions are not driven by this simplification.

4.5 Relation to the literature

- The paper extends the knowledge capital tradition of **Griliches**.
- Methodologically it builds on **Olley–Pakes**, **Levinsohn–Petrin**, and **Ackerberg–Caves–Frazer**, but replaces nonparametric inversion of the dynamic problem with a **parametric inversion of static input demand**.
- It also connects to **Hall–Hayashi** and **Klette** by allowing uncertainty and nonlinearity in the R&D process, but goes further by recovering firm-level productivity paths.

5 Tables and figures

5.1 Table 1: sample structure and R&D heterogeneity

- Output growth is strong across industries, roughly from **2% to almost 9%** per year, while capital growth is also high and labour growth is weak.
- Firms differ sharply in innovative intensity. High-tech industries such as chemicals, machinery, and transport have performer shares around **55%–57%**; low-tech industries such as timber/furniture and paper/printing have performer shares around **13%–17%**.
- Stable performers are especially concentrated in the more innovative industries.
- **Read:** The sample has enough cross-industry heterogeneity in R&D behaviour to discipline a model in which the productivity process depends on R&D.

TABLE 1
Descriptive statistics

Industry	Obs. ^a	Firms ^b	R&D ^c (%)	Rate of growth ^d					Non-R&D ^e			
				Output (%)	Capital (%)	Labour (%)	Materials (%)	Price (%)	Obs. (%)	Firms (%)	Output (%)	R&D share (%)
1. Metals and metal products	105	38	1.1	0.005	0.006	0.011	0.012	0.01	43	43	72	0.018
2. Non-metallic minerals	402	151	0.1	0.018	0.018	0.018	0.018	0.018	151	151	74	0.018
3. Chemical products	104	19	1.1	0.018	0.018	0.018	0.018	0.018	19	19	74	0.018
4. Agric. and food, machinery	176	107	1.1	0.018	0.018	0.018	0.018	0.018	107	107	74	0.018
5. Transport equipment	107	107	1.1	0.018	0.018	0.018	0.018	0.018	107	107	74	0.018
6. Textiles, leather, and shoes	107	107	1.1	0.018	0.018	0.018	0.018	0.018	107	107	74	0.018
7. Food, drink, and tobacco	107	107	1.1	0.018	0.018	0.018	0.018	0.018	107	107	74	0.018
8. Timber, furniture, and others	107	107	1.1	0.018	0.018	0.018	0.018	0.018	107	107	74	0.018
9. Paper and printing products	107	107	1.1	0.018	0.018	0.018	0.018	0.018	107	107	74	0.018

^a Compares for 1980 to 1999 excluding the first observation for each firm.

^b Compares for 1980 to 1999 excluding the first observation for each firm.

TABLE 2
Production function estimates and specification tests

Industry	OLS ^a				GMM ^b				Overidentifying restrictions				Parameter restrictions			
	Capital (std. err.)	Labour (std. err.)	Materials (std. err.)	Price (std. err.)	Capital (std. err.)	Labour (std. err.)	Materials (std. err.)	Price (std. err.)	$\chi^2(4)$	p -val	$\chi^2(5)$	p -val	$\chi^2(4)$	p -val	$\chi^2(5)$	p -val
1. Metals and metal products	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
2. Non-metallic minerals	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
3. Chemical products	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
4. Agric. and food, machinery	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
5. Transport equipment	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
6. Textiles, leather, and shoes	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
7. Food, drink, and tobacco	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
8. Timber, furniture, and others	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
9. Paper and printing products	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001

^a All standard errors are robust to heteroskedasticity and autocorrelation.

TABLE 3
Production function estimates and specification tests: alternative materials demand function

Industry	GMM ^a				Overidentifying restrictions				Parameter restrictions				Stability				Degree of non-stochasticity			
	Capital (std. err.)	Labour (std. err.)	Materials (std. err.)	Price (std. err.)	$\chi^2(4)$	p -val	$\chi^2(5)$	p -val	$\chi^2(4)$	p -val	$\chi^2(5)$	p -val	$\chi^2(4)$	p -val	$\chi^2(5)$	p -val	$\chi^2(4)$	p -val	$\chi^2(5)$	p -val
1. Metals and metal products	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
2. Non-metallic minerals	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
3. Chemical products	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
4. Agric. and food, machinery	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
5. Transport equipment	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
6. Textiles, leather, and shoes	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
7. Food, drink, and tobacco	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
8. Timber, furniture, and others	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
9. Paper and printing products	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001

^a All standard errors are robust to heteroskedasticity and autocorrelation.

5.2 Table 2 and Table 3: production-function estimates and inversion choice

- With the preferred labour inversion, the GMM capital coefficient ranges roughly from **0.06 to 0.23**, labour from **0.11 to 0.31**, and materials from **0.59 to 0.77**.
- The overidentifying restrictions are not rejected in any industry at the 5% level.

- When productivity is recovered using materials instead of labour, estimates become less stable and specification problems become more pronounced.
- **Read:** Labour inversion is the empirically preferred route. The production function looks economically sensible, while the materials inversion appears more vulnerable to measurement problems.

5.3 Table 4: permanent versus temporary labour

- Temporary workers earn lower wages than permanent workers: the estimated temporary-worker wage coefficient is negative in every industry.
- The implied labour-quality parameter α typically lies between about **0.66 and 0.88**, indicating meaningful productivity differences between permanent and temporary labour.
- Yet the production-function estimates after correcting for this distinction remain close to the baseline.
- **Read:** Labour-market institutions matter in Spain, but the core R&D–productivity conclusions are robust to modelling this extra margin.

TABLE 4
Production function estimates, specification tests, and productivity levels: permanent and temporary workers

Industry	Wage regression ^a				α^b		GMM ^b		Overidentif. tests ^a		Diff. of means ^b	
	Temp. (s.e.)	R ²	Obs. ^b	Firms ^b	Mean (s.e.)	Caps. (s.e.)	Lab. (s.e.)	Mat. (s.e.)	$\chi^2(df)$	p val.	≤ 200	> 200
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
1. Metals and metal products	-0.405 (0.063)	0.599	1046	262	0.786 (0.200)	0.094 (0.009)	0.173 (0.015)	0.692 (0.012)	67.916 (51)	0.057	0.010	0.003
2. Non-metallic minerals	-0.414 (0.105)	0.701	477	129	0.827 (0.169)	0.067 (0.012)	0.209 (0.014)	0.706 (0.012)	53.832 (47)	0.229	0.050	0.045
3. Chemical products	-0.484 (0.086)	0.680	1019	250	0.852 (0.165)	0.055 (0.010)	0.196 (0.016)	0.741 (0.012)	55.434 (47)	0.187	0.025	0.041
4. Agric. and ind. machinery	-0.182 (0.088)	0.594	439	118	0.817 (0.188)	0.046 (0.015)	0.247 (0.020)	0.691 (0.013)	44.974 (44)	0.431	-0.060	0.024
6. Transport equipment	-0.825 (0.072)	0.664	538	138	0.884 (0.135)	0.079 (0.012)	0.208 (0.018)	0.673 (0.016)	42.877 (47)	0.644	0.060	0.065
7. Food, drink, and tobacco	-0.366 (0.064)	0.659	1222	290	0.732 (0.230)	0.081 (0.011)	0.159 (0.015)	0.756 (0.011)	67.638 (46)	0.021	-0.011	0.015
8. Textile, leather, and shoes	-0.323 (0.055)	0.620	1023	269	0.703 (0.260)	0.065 (0.011)	0.240 (0.017)	0.640 (0.014)	61.745 (51)	0.144	0.008	0.016
9. Timber and furniture ^c	-0.323 (0.070)	0.618	499	129	0.662 (0.222)	0.128 (0.028)	0.121 (0.011)	0.728 (0.010)	-	-	-0.060	0.060
10. Paper and printing products	-0.259 (0.107)	0.694	518	142	0.805 (0.182)	0.131 (0.015)	0.223 (0.022)	0.624 (0.019)	44.424 (42)	0.370	-0.054	0.047

^a All standard errors are robust to heteroskedasticity and autocorrelation.
^b Conditional on nonzero temporary workers in period $t-1$.
^c First-step GMM estimates.

TABLE 5
Non-linearity and uncertainty and comparison to knowledge capital model

Industry	Exogeneity test		Separability test		$\text{Var}(\xi_{jt})$ $\text{Var}(\omega_{jt})$	Knowledge capital model tests					
	$\chi^2(10)$	p val.	$\chi^2(3)$	p val.		Basic		Generalization 1		Generalization 2	
	(1)	(2)	(3)	(4)	(5)	$N(0,1)$	p val.	$N(0,1)$	p val.	$N(0,1)$	p val.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
1. Metals and metal products	65.55	0.000	16.360	0.001	0.735	0.407	-2.815	0.002	-2.431	0.008	-1.987
2. Non-metallic minerals	92.65	0.000	13.027	0.005	0.842	0.410	-2.041	0.021	-1.541	0.062	-0.784
3. Chemical products	46.79	0.000	8.647	0.034	0.749	0.244	-3.239	0.001	-2.090	0.038	-1.400
4. Agric. and ind. machinery	51.88	0.000	11.605	0.009	1.410	0.505	-2.693	0.004	-1.588	0.056	-1.493
6. Transport equipment	56.85	0.000	18.940	0.000	1.626	0.524	-2.317	0.010	-2.082	0.021	-1.821
7. Food, drink, and tobacco	38.29	0.000	7.186	0.066	1.526	0.300	-3.263	0.001	-2.499	0.006	-0.901
8. Textile, leather, and shoes	29.91	0.001	18.417	0.000	1.121	0.750	-2.770	0.003	-1.788	0.077	-1.488
9. Timber and furniture	118.17	0.000	32.260	0.000	1.417	0.515	-2.510	0.006	-2.097	0.018	-1.028
10. Paper and printing products	39.73	0.000	23.249	0.000	0.713	0.433	-3.076	0.001	-2.210	0.014	-1.395

5.4 Table 5: exogeneity, nonlinearity, uncertainty, and knowledge-capital comparisons

- The exogenous Markov process is rejected in **every industry**.
- Separability between already attained productivity and current R&D is rejected in almost all industries.
- The ratio $\text{Var}(\xi_{jt})/\text{Var}(\omega_{jt})$ ranges from about **0.24 to 0.75**, meaning that the unpredictable component explains a large share of productivity variation.
- Simple and generalized knowledge capital models are also rejected by the data.
- **Read:** This is the core empirical table. It says productivity is **not exogenous**, the R&D process is **nonlinear**, and uncertainty is **first-order**.

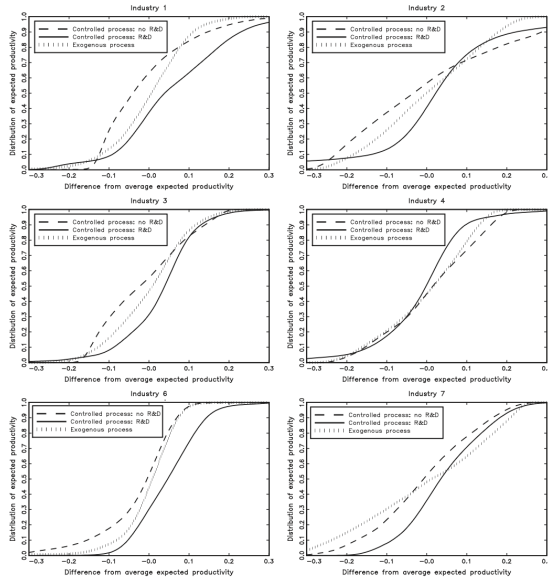


FIGURE 1
Distribution of expected productivity

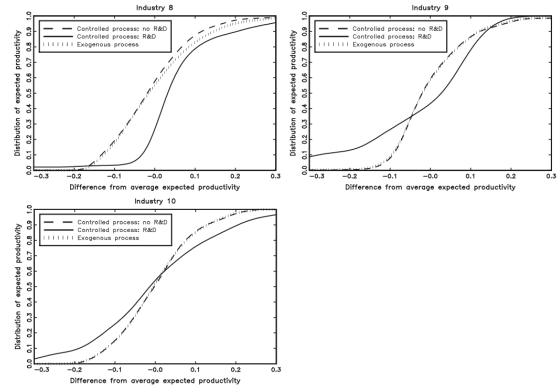


FIGURE 1
Continued

5.5 Figure 1, Table 6, and Figure 2: expected productivity of performers and non-performers

- In most industries with medium or high innovative activity, the distribution of expected productivity for performers lies to the **right** of that for non-performers.
- Differences in mean expected productivity are typically around **3% to 5%** in the more innovative industries.
- Once R&D is omitted from the productivity process, these differences blur sharply.
- **Read:** R&D-performing firms tend to occupy a better part of the productivity distribution, and modelling productivity as exogenous hides that fact.

TABLE 6
Productivity levels

Industry	Size	Diff. of means	Kolmogorov-Smirnov tests ^a						
			Mean with R&D is greater		Distributions are equal		Distribution with R&D dominates		
			<i>t</i>	<i>p</i> val.	<i>S</i> ₁	<i>p</i> val.	<i>S</i> ₂	<i>p</i> val.	
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	
1. Metals and metal products	≤ 200	0.051	-4.992	1.000	1.720	0.005	0.388	0.740	
	> 200	0.038	-2.920	0.998					
2. Non-metallic minerals	≤ 200	0.030	-1.320	0.905	1.167	0.131	0.938	0.172	
	> 200	0.025	-1.313	0.904					
3. Chemical products	≤ 200	0.033	-4.531	1.000	1.440	0.032	0.351	0.782	
	> 200	0.028	-2.003	0.975					
4. Agric. and ind. machinery	≤ 200	-0.023	1.807	0.036	0.988	0.283	0.988	0.142	
	> 200	-0.077	3.644	0.000					
6. Transport equipment	≤ 200	0.073	-6.377	1.000	1.770	0.004	0.000	1.000	
	> 200	0.038	-4.126	1.000					
7. Food, drink, and tobacco	≤ 200	0.034	-3.476	1.000	1.708	0.006	0.198	0.925	
	> 200	0.028	-3.073	0.999	0.610	0.850	0.227	0.902	
8. Textile, leather, and shoes	≤ 200	0.044	-2.543	0.994	3.173	0.000	0.335	0.799	
	> 200	0.022	-1.552	0.939	0.511	0.957	0.426	0.696	
9. Timber and furniture	≤ 200	-0.020	0.724	0.237					
	> 200	0.014	-0.610	0.721					
10. Paper and printing products	≤ 200	-0.013	0.590	0.279	0.930	0.353	0.709	0.366	
	> 200	0.028	-1.400	0.917					

^a Applied to a firm's average expected productivity when each sample has more than 20 firms.

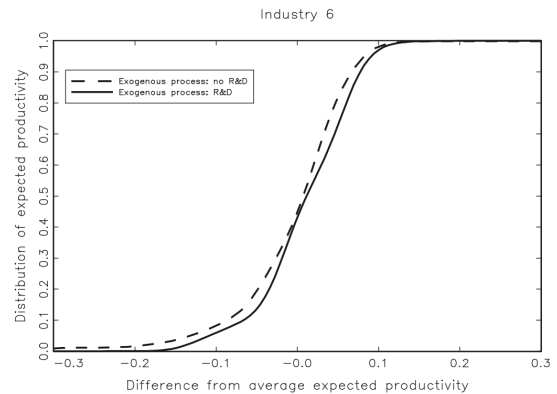


FIGURE 2
Distribution of expected productivity. Exogenous Markov process

5.6 Table 7 and Figure 3: return to R&D and persistence in productivity

- The mean elasticity of output with respect to R&D expenditures is close to **0.015**, but varies from about **-0.006** to **0.046** across industries.
- Negative marginal returns appear in parts of the distribution, consistent with indivisibilities, strategic motives, or local boundary effects in flexible estimation.
- Productivity persistence is often **lower for performers** than for non-performers.
- Figure 3 shows that greater uncertainty is associated with **less persistence**.
- Relative to the knowledge capital model, this paper finds **lower persistence** and **more heterogeneity** in the effect of R&D.
- **Read:** R&D adds new knowledge, but it also makes productivity more volatile and less inertial than classic models imply.

TABLE 7
Elasticities of output with respect to R&D expenditures and already attained productivity

Industry	Elasticity wrt. R_{t-1} ^a										Knowledge capital model Elasticity wrt. C_t and R_{t-1} ^c			
	Elasticity wrt. y_{t-1} ^b										Gross output		Value added	
	Q_1	Q_2	Q_3	Mean	Q_1	Q_2	Q_3	Q_1	Q_2	Q_3	ϵ (s.e.)	Mean	ϵ (s.e.)	Mean
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
1. Metals and metal products	-0.013	0.007	0.021	0.022	0.504	0.619	0.755	0.441	0.759	0.901	0.003	0.001	0.025	0.005
2. Non-metallic minerals	-0.018	-0.012	0.000	-0.006	0.433	0.477	0.575	0.377	0.646	0.878	0.005	0.002	0.016	0.008
3. Chemical products	0.009	0.011	0.014	0.013	0.459	0.523	0.634	0.547	0.815	0.947	0.018	0.003	0.075	0.014
4. Agric. and ind. machinery	-0.017	-0.009	0.021	0.005	0.434	0.721	0.791	0.729	0.894	0.979	-0.003	-0.001	0.025	0.005
5. Transport equipment	-0.034	-0.008	0.010	0.020	0.404	0.615	0.727	0.423	0.513	0.646	0.009	0.002	0.017	0.004
6. Food, drink, and tobacco	-0.008	0.010	0.026	0.020	0.445	0.705	0.867	0.822	0.930	0.965	0.002	0.001	0.046	0.011
7. Textile, leather, and shoes	-0.003	0.014	0.051	0.046	0.990	0.525	0.626	0.491	0.805	0.689	0.011	0.002	0.018	0.003
8. Timber and furniture	-0.031	0.005	0.048	0.004	0.458	0.585	0.814	0.303	0.430	0.641	0.018	0.005	0.074	0.019
9. Paper and printing products	-0.036	0.022	0.049	0.013	0.405	0.676	0.812	0.569	0.644	0.670	0.013	0.004	0.041	0.012

^a We trim 2.5% of observations at each tail of the distribution of R_{t-1} .

^b We trim observations below zero and above unity.

^c All standard errors are robust to heteroskedasticity and autocorrelation.

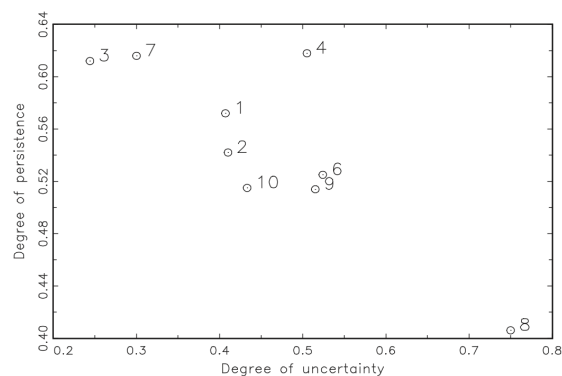


FIGURE 3
Persistence and uncertainty

5.7 Table 8 and Figure 4: productivity growth, rates of return, and uncertainty

- Productivity growth is highest in some of the high-R&D industries, reaching around **2% to 2.7%** in some cases.
- Firms that perform R&D account for about **65% to 90%** of productivity growth in the industries with medium or high innovative activity.
- The **gross rate of return to R&D** ranges roughly from **0.5 to 1.1**, while the **net rate** is often around **0.4** on average, with substantial cross-industry variation.
- The net return to R&D is about **1.2 to 4.4 times** the net return to physical capital.
- Figure 4 shows a positive relationship between the **net return to R&D** and the **degree of uncertainty** in the productivity process.
- **Read:** R&D has high returns partly because it is risky, and a substantial fraction of innovative spending goes to preserving existing productivity rather than only generating new breakthroughs.

TABLE 8
Productivity growth, rates of return to R&D and investment in physical capital, and degree of uncertainty

Industry	Rates of return									
	Productivity growth			R&D ^a		Capital	Ratio	Regr. of $\ln\left(\frac{t_j^2}{\text{Var}(y_j)}\right) \cdot \text{cap}^b$		Knowledge
	Total	R&D (%)	No R&D (%)	Gross rate	Net rate	Net rate		R&D (s. e.)	Inv. (s. e.)	eq. mod. Net rate (s. e.) ^c
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1. Metals and metal products	0.0172	0.0225 (73.0)	0.0099 (27.0)	1.092	0.659	0.432	0.193	3.4	0.362 (0.164)	-0.326 (0.302)
2. Non-metallic minerals	0.0048	0.0101 (90.3)	0.0018 (9.3)	0.919	0.632	0.287	0.311	2.0	0.673 (0.231)	-0.018 (0.279)
3. Chemical products	0.0172	0.0210 (83.0)	0.0095 (17.0)	0.507	0.347	0.160	0.199	1.7	0.612 (0.141)	0.014 (0.258)
4. Agric. and ind. machinery	0.0157	0.0137 (86.1)	0.0234 (36.9)	0.745	0.296	0.449	0.247	1.2	0.772 (0.197)	0.123 (0.286)
6. Transport equipment	0.0266	0.0281 (75.7)	0.0237 (24.3)	0.848	0.480	0.368	0.168	2.9	0.756 (0.207)	-0.225 (0.364)
7. Food, drink, and tobacco	0.0019	-0.0053 (11.0)	0.0054 (88.6)	0.584	0.098	0.485	0.072	1.4	1.337 (0.164)	0.795 (0.206)
8. Textiles, leather, and shoes	0.0108	0.0049 (12.4)	0.0146 (87.6)	0.918	0.332	0.585	0.075	4.4	1.587 (0.172)	0.681 (0.184)
9. Timber and furniture	0.0102	0.0386 (46.4)	0.0062 (53.6)	0.788	0.384	0.405	0.279	1.4	2.072 (0.382)	0.093 (0.212)
10. Paper and printing products	0.0089	0.0015 (11.0)	0.0110 (88.6)	1.049	0.444	0.605	0.158	2.8	1.284 (0.274)	-0.455 (0.333)

^a We calculate the first two terms of the decomposition in Equation (23) and infer the third term. We trim 5% of observations at each tail of the distribution of $\hat{\lambda}_{j-1}$. We further trim up to 30% of observations according to their rates.
^b All standard errors are robust to heteroskedasticity and autocorrelation.

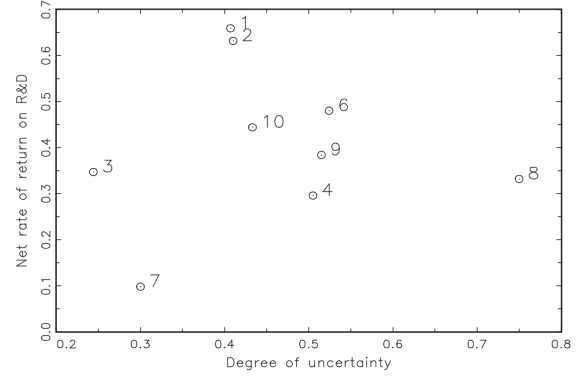


FIGURE 4
Return to R&D and uncertainty

6 Short summary

- **Method.** Model productivity as a controlled Markov process shifted by lagged R&D, and recover latent productivity from static input demands using a parametric inversion.
- **Identification.** Lagged input-demand objects and lagged R&D identify the law of motion for productivity, while firm-level wage and price variation are crucial for the inversion.
- **Data.** Unbalanced panel of 1,870 Spanish manufacturing firms in 9 industries during the 1990s, with unusually rich information on output, prices, wages, materials, investment, and R&D.
- **Main result.** The R&D process is uncertain, nonlinear, and heterogeneous. R&D performers tend to have higher expected productivity, contribute disproportionately to productivity growth, and earn net returns to R&D that often exceed the return to physical capital.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper provides a way to estimate production functions and the productivity process when productivity is **endogenous to R&D investment**.
- It shows that the link between R&D and productivity cannot be adequately represented by a simple knowledge-capital stock with constant depreciation.
- In Spanish manufacturing, R&D is a central determinant of both **cross-firm productivity differences** and **firm-level productivity growth**.

7.2 Economic intuition

- R&D does not map one-for-one into a deterministic stock of knowledge.
- Some R&D succeeds, some fails, and the payoff depends on what the firm already knows.
- Firms therefore invest not only to create new knowledge but also to **maintain** or **renew** existing productivity in the face of obsolescence and uncertainty.

- This makes productivity growth an uneven and heterogeneous process, even among firms with similar observed R&D spending histories.

7.3 Takeaway

This paper turns the R&D–productivity relationship from a simple accounting identity into a **dynamic stochastic process**. The main lesson is that R&D matters a lot, but it matters through a mechanism that is **state dependent**, **uncertain**, and **firm specific**. Any empirical approach that treats productivity as exogenous or knowledge accumulation as deterministic risks missing the central economics of innovation.